Solution

This can be written as $\left(3x^3-2x^2-7x+6\right)-\left(x^3-x^2+x+3\right)$

When there is a –ve sign before the brackets, it can be removed by changing the sign of every term inside the bracket.

$$\begin{aligned}\left(3x^3-2x^2-7x+6\right)-\left(x^3-x^2+x+3\right) &= 3x^3-2x^2-7x+6-x^3+x^2-x-3\\ &=\left(3x^3-x^3\right)+\left(-2x^2+x^2\right)+\left(-7x-x\right)+\left(6-3\right)\\ &=x^3\left(3-1\right)+x^2\left(-2+1\right)+x\left(-7-1\right)+\left(6-3\right)\\ &=2x^3-x^2-8x+3\end{aligned}$$

Hence, the degree of the expression is 3.

Example 3.17 Simplify and find the degree of the expression

$$\left(4m^2+3n\right)-\left(3m+9n^2\right)-\left(3m^2-6n^2\right)+\left(5m-n\right)$$

Solution

$$\begin{aligned}&\left(4m^2+3n\right)-\left(3m+9n^2\right)-\left(3m^2-6n^2\right)+\left(5m-n\right)\\ &= 4m^2+3n-3m-9n^2-3m^2+6n^2+5m-n\\ &=\left(4m^2-3m^2\right)+\left(3n-n\right)+\left(-3m+5m\right)+\left(-9n^2+6n^2\right)\\ &=m^2+2n+2m-3n^2\end{aligned}$$

Hence, the degree of the expression is 2.

Exercise 3.3

1. Fill in the blanks.
   (i) The degree of the term $a^3b^2c^4d^2$ is _____.
   (ii) Degree of the constant term is _____.
   (iii) The coefficient of leading term of the expression $3z^2y+2x-3$ is _____.
2. Say True or False.
   (i) The degree of m^2n and mn^2 are equal.
   (ii) $7a^2b$ and $-7ab^2$ are like terms.
   (iii) The degree of the expression $-4x^2yz$ is -4.
   (iv) Any integer can be the degree of the expression.
3. Find the degree of the following terms.
   (i) $5x^2$ (ii) $-7ab$ (iii) $12pq^2r^2$ (iv) -125 (v) $3z$
4. Find the degree of the following expressions.
   (i) x^3-1 (ii) $3x^2+2x+1$ (iii) $3t^4-5st^2+7s^3t^2$
   (iv) $5-9y+15y^2-6y^3$ (v) $u^5+u^4v+u^3v^2+u^2v^3+uv^4$

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